

Service-Oriented Enterprise Network Performance Analysis*

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Abstract: The service-oriented architecture (SOA) and the model-driven architecture (MDA) have been recognized as major evolutionary steps in enterprise integration (EI) in service-oriented computing environments. Service-oriented enterprise (SOE) networks (SOEN) are emerging with the significant advances of EI, SOA, and MDA. However, the implementation and optimization of SOEN is still lacking integrated SOA, MDA, and performance analysis and optimization (PAO) methods. This paper introduces an integrated solution of SOA and MDA with a simulation-based three-stage PAO method with stage 1 being an analytic hierarchy process (AHP)-based comprehensive performance calculation for service matching and binding, stage 2 being a simulation-based comprehensive performance evaluation for business process/service composition, and stage 3 being a business process simulation-based performance optimization for SOEN. The SOE architecture, performance analysis framework, performance indicators, and performance operators are discussed. The system uses MDA as the system development philosophy, SOA as the system implementation infrastructure, and the simulation-based PAO methods to analyze and optimize the SOEN performance. A case study of the SOEN illustrates the usage of the integrated solution.

Key words: service-oriented architecture (SOA); service-oriented enterprise (SOE); service-oriented business process (SOBP); performance analysis; simulation optimization; analytic hierarchy process (AHP)

Introduction

Information technologies (IT) such as the model-driven architecture (MDA), service-oriented architecture (SOA), and business process management (BPM) have facilitated business globalization, digitization, virtualization, and agility. An enterprise's competency now depends on the application and integration of human resources, advanced manufacturing technologies, IT technologies, management methods, and collaboration with other partners.

However, traditional enterprise integration (EI) suffers from (1) the lack of consistency between enterprise strategies, business processes, application systems, and IT infrastructures; (2) the implementation of inaccurate and inflexible business processes supported by the application systems; (3) the existence of large heterogeneities in the enterprise application systems and the lack of adaptability of the IT architecture; and (4) the lack of performance analysis and optimization applications for the integrated enterprise networks.

The SOA provides broad insight for tackling EI problems which enables enterprises to dynamically discover, aggregate, and reconfigure a range of web services through the Internet. SOA also offers mechanisms to dynamically integrate different technologies independent of the system's platform.

However, there are still some crucial problems

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related to the implementation and optimization of service-oriented enterprises (SOE) and SOE networks (SOEN). For example, there are only a limited number of the enterprise architectures, business processes and services, application system architecture, and performance analysis frameworks for the service-oriented computing environments. In addition, development methodologies, performance analyses, and optimization methods (for dynamic binding, composing, and usage) of SOEN are relatively scarce and not very systematic.

This paper presents an integrated implementation, performance analysis, and optimization method for SOEN to support full-scale, dynamic integration of SOEN. The system uses MDA as the system development methodology to achieve integration at the model level, SOA to construct the integration platform and implementation methods for various models, and business process simulations (BPS) to analyze and optimize the SOEN during its design, development, and implementation.

1 Related Works

An enterprise is an organization set that is gathered together to fulfill a unified set of objectives. A service system is an integrated system designed to enhance its efficiency, effectiveness, and adaptiveness with the combination of people (characterized by behaviors, attributes, values, etc.), processes (characterized by collaboration, customization, etc.), and products (characterized by software, hardware, infrastructures, etc.)^[1].

EI breaks down the organizational barriers to improve the synergy within the enterprise so that business goals can be achieved in a more productive and efficient way^[2]. EI can be classified into six categories according to the level of integration, i.e., the organization integration^[3], the process integration^[4], the data integration^[5], the application integration^[2], the service integration^[6], and the semantic integration^[7].

To analyze an enterprise with a wide range of roles and content, EI and enterprise architecture (EA) studies usually investigate an enterprise in terms of different domains, different abstract layers, and different perspectives. Many enterprise frameworks and architectures have been proposed, e.g., the Zachman framework^[8], the computer-integrated manufacturing (CIM)-open system architecture (CIM-OSA)^[9], the

CIM flow-integrated enterprise modeling architecture and method (CIMFlow-IEMAM)^[10], the architecture of integrated information systems (ARIS)^[11], the Purdue enterprise reference architecture (PERA)^[12], the graph with results and activities interrelated (GRAI) method and GRAI integrated methodology (GIM)^[13], the ICAM (integrated computer-aided manufacturing) definition method (IDEF)^[14], the integrated enterprise modeling (IEM) method^[15], and the supply-chain operation reference-model (SCOR)^[16].

SOE performance analysis and optimization involve all the abstract levels and aspects of an enterprise, especially service-oriented business processes (SOBP) which integrate services, activities, resources, enterprise partners, etc. The three main levels for SOBP performance analysis and optimization are service matching and binding of an abstract service with its physical services, service orchestration centered on a specific service, and service choreography from a global perspective involving many services and their environments. All SOBP should be analyzed and optimized before being actually implemented.

There are three main types of performance analysis methods, i.e., model analysis methods^[17], data analysis methods^[18,19], and simulation-based methods.

Business process simulations (BPS) simulate the dynamic behavior of processes through a simulation engine driven by discrete events, record the relevant data during the simulation, and analyze the key performance indexes and bottlenecks by the data. Nowadays, BPS could use the special simulation tools developed for business processes^[20], Petri-net tools^[21], or general discrete-event simulators^[22]. BPS research on service-oriented environments is a new topic emerging in recent years. Zheng et al.^[23] proposed an interactive-event-based workflow simulation in a service-oriented environment. Tsai et al.^[24] used modeling and simulation for service-oriented software development. Chen extended the Petri-net, communicating sequential processes (CSP), and quality of service (QoS)-aware methods and technologies to BPS for processes/services composition in a service-oriented environment^[25].

2 Service-Oriented Enterprises

2.1 Architecture of service-oriented enterprises

The service-oriented enterprise architecture (SOEA) is

a development and operational architecture for enterprise integration in a service-oriented computing environment. SOEA provides an integrated development and execution environment based on SOA and MDA as shown in Fig. 1. The main components of SOEA are as follows:

(1) The enterprise object refers to the elements involved in EI. All the elements defined in the EI model can be the enterprise objects of SOEA, such as enterprise systems (e.g., the enterprise resource plan system and the supply-chain management system), business processes, organization units, and data. They must be wrapped by service adapters before being integrated into SOEA.

(2) The service adapter converts the access interface of the enterprise object into a standard service interface. An adapter works as an intermediary between the service and the enterprise object. It models the data and functionality in the enterprise object and provides a service interface to access the functionality.

(3) The service repository stores the models and interface descriptions in a central location that is accessible to the enterprise objects. A repository permits

searching for models and interface descriptions and maps abstract service identifiers with physical addresses.

(4) The service bus is a standards-based communication layer in an SOA that enables services to be used across multiple communication protocols. The service bus provides an intermediate approach which abstracts the interaction details between the service consumer and the service producer. It simplifies service deployment and management and promotes service reuse in a heterogeneous environment.

(5) The service infrastructure is a unified approach to service management that combines the capabilities of the enterprise service bus and business process management. The service infrastructure creates a fabric of services for constructing and deploying new composite business applications. The main functions of the service infrastructure include service registry, discovery, composition, orchestration, choreography, service data management, and service lifecycle management.

(6) The enterprise integration model defines various aspects of the business model integration, i.e., the organization, process, data, system, and service layers.

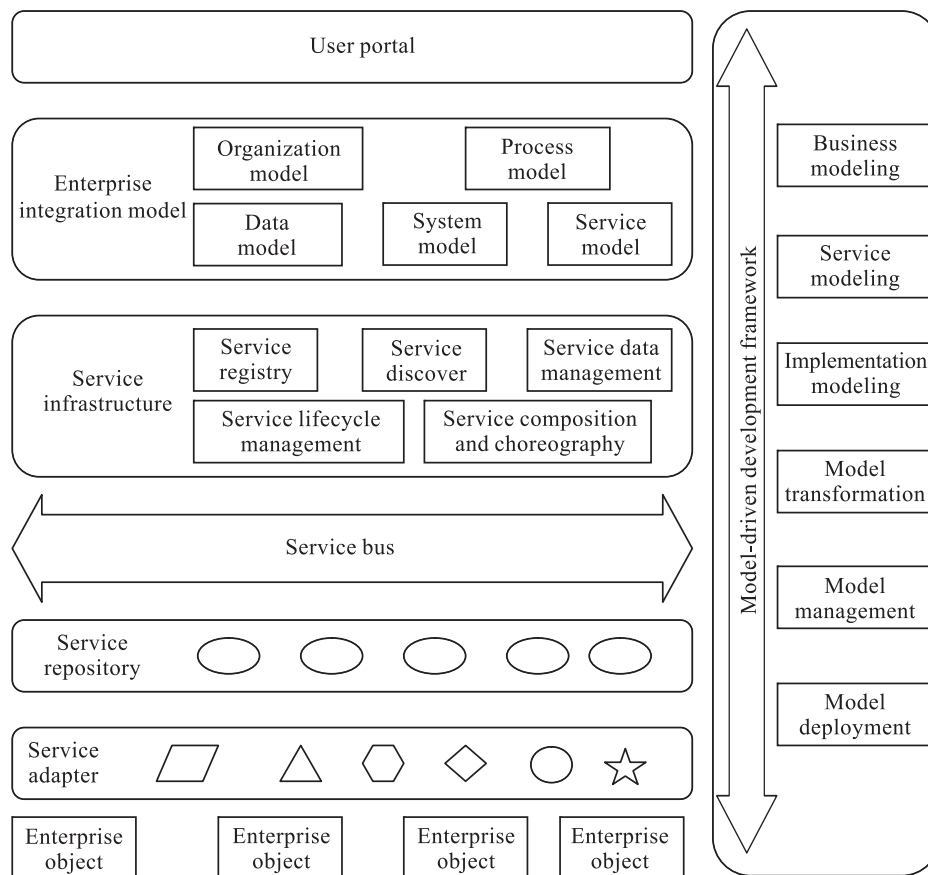


Fig. 1 Service-oriented enterprise architecture

Integration of the different business models enables accurate modeling of an actual scenario. Eligible services can be matched, composed, and executed through the mapping of the computer independent model (CIM) to platform independent model (PIM), and PIM to platform-specific model (PSM).

(7) The user portal is the front end of the integration platform which supports dynamic workplaces and working environments tailored to a certain purpose. It generates a dynamic, configurable user interface which enables the user to transparently access the platform functions and back end applications to obtain the correct information the right time.

(8) The model-driven development framework crosses all the SOEA levels and provides various MDA tools for the development of EI systems. It includes business modeling, service modeling, implementation modeling, and model transformation, management, and deployment.

2.2 Service-oriented business process

Business processes supported by SOEA include not only all traditional process elements but also networked services (NS). Thus, an SOBP may comprise the following three types of activities:

(1) The manual-activity means an activity implemented by human beings. People can be helped by resources including enterprise applications and services.

(2) The NS-activity represents an activity implemented by the NS which is normally called a service.

(3) If an activity is implemented by an enterprise application system that is not an NS, then the activity is an application-system-activity.

Since each NS implements some business functions as an application and is exposed to and used by the Internet, its performance has three main level attributes as follows:

(1) Business-level attributes, where the NS is considered an activity that fulfills the business operation and strategy objectives. They describe business functions and the business performance of an activity.

(2) Application-system-level attributes, where the NS is considered an application system that implements the business functions of an activity. They describe the activity's relationship to the application system and its IT application system performance from the IT perspective when an abstract service (ASV) is matched to some physical services (PSV) or a

process composed of PSV that are matched and bound to the system, to application-system-activities, or to manual-activities.

(3) IT-infrastructure-level attributes, where the NS is considered to be a concrete computer system so the attributes describe the relations between the IT functions and of the system performance with the IT infrastructure (e.g., software, hardware, and networks).

3 Simulation-Based SOEN Performance Analysis and Optimization

The SOEN performance analysis and optimization involve multiple enterprise layers, multiple organizations, multiple indicators, and multiple stakeholders. The subjective and objective factors can be quantified using simulations combined with the analytic hierarchy process (AHP) method.

3.1 Framework for SOEN performance analysis and optimization

The framework for SOEN performance analysis and optimization layers as shown in Fig. 2 has four main layers as follows:

(1) The enterprise layer uses enterprise modeling (EM)/EI technologies to achieve the SOEN integration and modeling (intra- or inter-organization/enterprise) including the business processes/services (BP/S). The BP/S is the core of the integrated enterprise physical, social, and knowledge strata. The physical stratum describes enterprise products, functions, organizations, resources, and information. The social stratum involves the relationships between an enterprise and other enterprises including the time, causal, communication, control, and coordination relationships. The knowledge stratum includes the business rules, exception handling rules, resources scheduling rules, event handling rules, etc.

(2) The BP/S layer builds a process/service model with modeling tools and stores the models into the model database (DB) for process execution and simulations. The BP/S execution is accompanied by service matching and composition which is supported by the SOEA. The simulation scenario configurations comply with the process decision making supported by the performance analysis applications and analysts' knowledge.

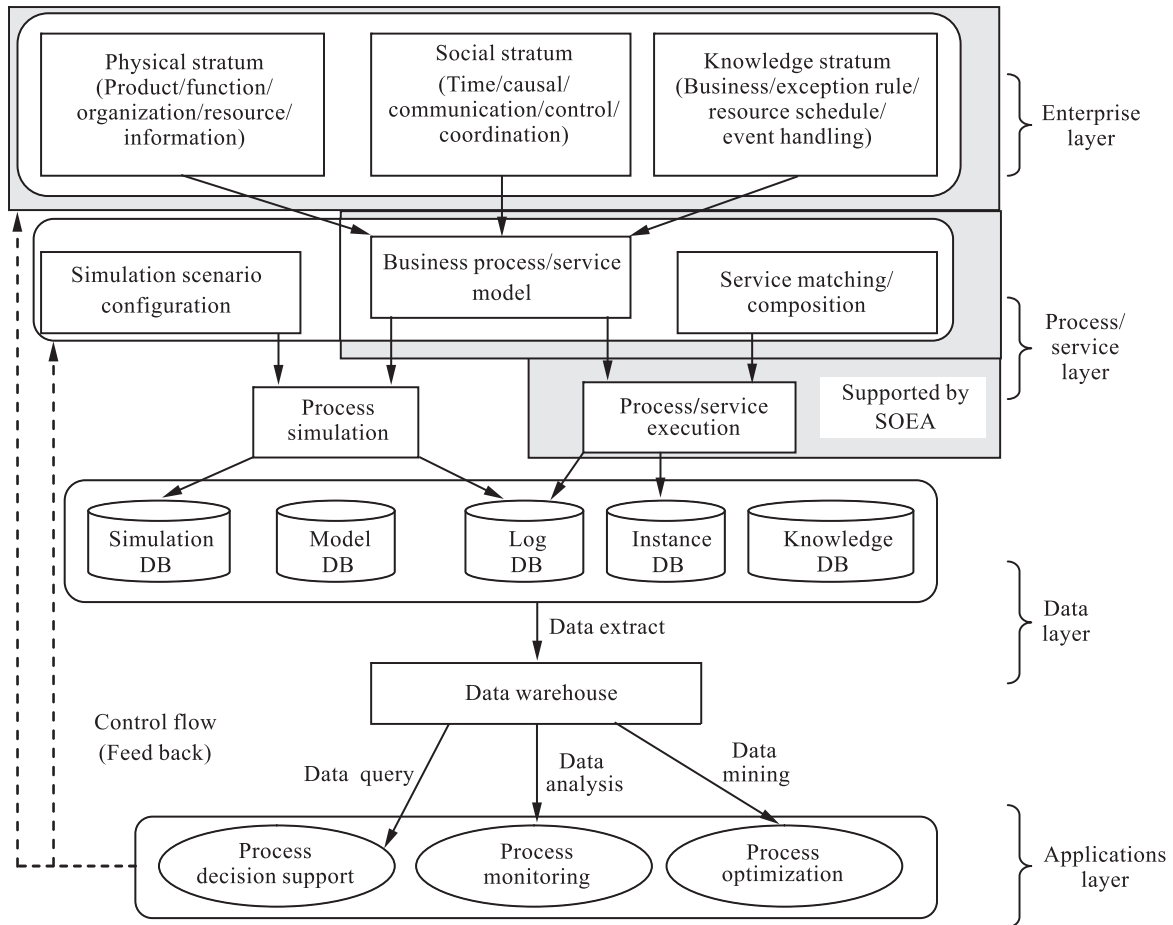


Fig. 2 SOEN performance analysis and optimization framework

(3) The data layer includes the model DB, instance DB, log DB, simulation DB, knowledge DB, and other DB. The BP/S execution data and log generated during the process/service executing are stored into the instance DB and log DB, while the simulation data and log are stored in the simulation DB and the log DB. All the data is source data for data analysis and mining. The original data can be extracted by the ETL (extract, transform, and load) tools and stored into the data warehouse.

(4) The performance analysis application layer contains three application systems supporting the process performance monitoring, decision-making, and optimization. The applications results are fed back to the enterprise and BP/S layers to adjust the enterprise, BP/S models, or simulation scenario.

3.2 SOEN performance indicators

Section 2 shows that the SOEN performance indicators are mainly determined by the NS performance attributes. The indicators are selected subject to the

analysis objectives. The SOEN (or NS) indicators selected here can be extended with others at each level, but their structure is stable. The three-level NS indicators are closely related since the IT-infrastructure-level indicators are the foundations of the application-level and the business-level indicators. The application-level indicators are the basis for the business-level indicators.

(1) Business-level indicators

• Business reliability (R_b)

R_b denotes the reliability of the business performance promised by the service providers. $R_b \in (0, 1]$ and $R_b = 1 - \left(\sum_{i=1}^n |x_i - y_i| / x_i \right) / n$, where x_i is the promised value of the indicator provided by the service providers and y_i is the real value of the indicator ascertained from the assessments provided by the n consumers.

• Business time (T_b)

T_b denotes the completion time for a business process promised by the service providers. It is issued by the providers and may range from 2 d to 5 d with a

specified randomness.

- Business cost (C_b)

C_b denotes the business cost promised by the service providers in a range such as \$200-500 with a specified randomness.

- Business flexibility (F_b)

$F_b = F_f \otimes F_p$ denotes the business flexibility, where F_f denotes the number of NS optional implementation schemes with the same function, e.g., a function named TRANSPORTATION may be implemented by a train, by a car, or by a plane, then $F_f = 3$; and F_p denotes the number of optional performance schemes for a certain function scheme, e.g., the function named BY TRAIN may have the performance schemes: 10 days, \$100 or 1 day, \$500, thus $F_p = 2$. Symbol “ $\otimes$ ” here indicates an operator and its end is the combinatory number of the function schemes and the performance schemes.

- Business organization relationship (O_b)

$O_b \in (0,1]$ denotes the business organization relationship between the SOBP and its service providers at the business level. It is used to weigh the correlations for the different enterprises at the business level.

(2) Application-level indicators

Application-level indicators denote the system qualities of the NS.

- System response time (T_q)

$T_q = \sum_{i=1}^n T_{q,i} / n$ denotes the time span between the time when the application system receives an input and the time when it outputs the corresponding outcomes. T_q can be predicted from existing records.

- System availability (A_q)

$A_q = T_s / T_t$ denotes the time ratio during which a service can be successfully available (T_s) in a specified interval (T_t). “Success” here means that the service will provide a correct output in an expected interval.

- System organization relationship (R_q)

$R_q \in (0,1]$ denotes the organizational relation between the SOBP and its service providers from the IT perspective. It can be used to weigh the IT collaboration relations between the different enterprises, application systems.

- System flexibility (F_q)

$F_q = f(F_b)$ denotes the application system’s flexibility that is related to the implementation technologies for a service such as its function and interface descriptions. It is the technical base of the indicator F_b .

- System throughput (P_q)

$P_q = N / T_t$ denotes the transactions handled by a service per unit time.

(3) IT-infrastructure-level indicators

- IT component reliability (R_{IT})

$R_{IT} = T_w / T_t$ denotes the ratio of the time span for a component to be working correctly (T_w) to a given interval (T_t).

- IT resources utilization (U_{IT})

$U_{IT} = T_u / T_t$ denotes the ratio of the time that a component is being used (T_u) to a given interval (T_t).

- System configuration and structure (C_{IT})

$C_{IT} \in (0,1]$ denotes the comprehensive evaluation of the system’s structures, scheduling policies, work patterns, etc. It can be analyzed by quality analysis methods, benchmarking, etc.

The SOEN (or SOBP) performance indicators may cover the five NS business-level indicators, the business time (T_b), cost (C_b), reliability (R_b), flexibility (F_b), and organizational relations (O_b).

3.3 Performance operators

The SOBP includes four basic structures, i.e., sequential, concurrent, alternative with probabilities, and iterative structures. The performance indicators for each activity (or service) and the performance calculation operators (Φ_k) for the four structures are used to calculate the SOBP’s overall performance. Φ_k is defined as below to calculate the key SOBP performance indicators, where q_i denotes the performance indicator i of an activity, w_i is the weight, and n is the number of activities in the SOBP.

- (1) Linear operators, $L(q_1, q_2, \dots, q_n) = \sum_{i=1}^n w_i q_i$. They

mainly include

- Summation operators, $L_{\text{sum}}(q_1, q_2, \dots, q_n) = \sum_{i=1}^n q_i$,

which can be used to calculate the costs or times for multiple processes or activities, where $w_i = 1$, $1 \leq i \leq n$.

- Average operators, $L_{\text{avg}}(q_1, q_2, \dots, q_n) = \frac{1}{n} \sum_{i=1}^n q_i$,

which can be used to calculate the reliability and availability for multiple processes or activities, where $w_i = 1/n$, $1 \leq i \leq n$.

- Probability operators, $L_{\text{prb}}(q_1, q_2, \dots, q_n) = \sum_{i=1}^n p_i q_i$,

which can be used to analyze multiple processes or activities with alternative structures, where $w_i = p_i$ is the probability that the activity may be executed, $1 \leq i \leq n$.

- Numeral multiplication operators, $L_{\text{tim}}(q) = kq$, which are usually used to analyze multiple processes or activities with iterative structures, where k is the number of iterations.

(2) Extremum value operators, $\overline{M}(q_1, q_2, \dots, q_n) = \max_{i=1}^n q_i$ or $\underline{M}(q_1, q_2, \dots, q_n) = \min_{i=1}^n q_i$ are usually used to analyze multiple processes or activities with concurrent structures.

(3) Multiplication operators, $T(q_1, q_2, \dots, q_n) = \prod_{i=1}^n q_i$

are used to calculate the availability of multiple processes or activities with sequential structures.

(4) Power operators, $P(q) = q_i^k$ are usually used to analyze multiple processes or activities with iterative structures, where k is the number of iterations.

Almost all the calculations can be denoted by those operators. For example, typical performance indicators^[26,27] can be denoted in Table 1.

Table 1 Performance operators for various business process/service structures

Performance indicators	Cost	Time	Reputation	Success	Availability
Sequential	L_{sum}	L_{sum}	L_{avg}	T	T
Alternative	L_{prb}	L_{prb}	L_{prb}	L_{prb}	L_{prb}
Concurrent	L_{sum}	$\overline{M}$	L_{avg}	$\underline{M}$	$\underline{M}$
Iterative	L_{tim}	L_{tim}	L_{avg}	P	P

3.4 Performance calculation and optimization

Stage 1 Analytic hierarchy process (AHP)-based comprehensive performance calculation for service matching and binding

Step 1 Determine the performance indicator set.

Each indicator in Section 3.2 can be assigned a reference number k and a corresponding weight w_k , $k=1, \dots, K$, where K is the total number of indicators selected by the users.

Assume that an SOBP has n activities, m of them are NS-activities, and each ASV has J_i PSV. Thus, the performance of the i -th ASV, P_i , is a two-dimensional matrix as

$$P_i = (P_{i,j,k}; 1 \leq j \leq J_i, 1 \leq k \leq K) \quad (1)$$

Consequently, the performance of the j -th PSV, $P_{i,j}$,

is a vector

$$P_{i,j} = (P_{i,j,k}; 1 \leq k \leq K) \quad (2)$$

Similarly, the performance of the manual activities and application system activities can be denoted as a vector too:

$$P_i = (P_{i,k}; 1 \leq k \leq \overline{K}) \quad (3)$$

where $\overline{K}$ is the number of performance indicators for the manual activity or the application system activity.

Step 2 Determine the weights of all indicators based on AHP.

Compare the indicators' importance one by one and give a value for each indicator, and then solve the judgment matrix A to get the weight set W . If the calculation consistency evaluation of W can be verified, this W is the needed weights vector.

Step 3 Determine the fuzzy relation and fuzzy comprehensive evaluation model.

Firstly, define a fuzzy remarks set V with elements v_l , $l = 1, \dots, m$. Then, the remarks of each indicator in Eq. (2) can be denoted as r_{kl} . The fuzzy relation matrix, R , derived from $P_{i,j}$ to V is

$$R = \{r_{kl}, 1 \leq k \leq K, 1 \leq l \leq m\} \quad (4)$$

According to the value of $B = WR = \{b_1, \dots, b_m\}$, we can arrange the order of all the ASV's PSVs. If only some PSV, U_i , with better performance are selected as candidates for the i -th ASV, then the SOBP has N potential composition schemes, where N is

$$N = \prod_{i=1}^m U_i \quad (5)$$

Stage 2 Simulation-based comprehensive performance evaluation for the business process/service composition

Some indicators such as R_b , F_b , and O_b describe the performance of the SOBP's structure, so they can be calculated analytically. However, T_b and C_b of each activity usually are not determinate but random variables with specified distributions which may be affected by many factors. Here, the system simulation method is used to calculate those indicators.

The SOBP simulation model will be iteratively executed by the simulation engine. During the simulations, the simulation engine schedules the enterprise organization, resources, employees, business rules, data, etc., and records all the predefined parameters. The optimization tools are used to analyze the performance for various configurations to determine the optimal SOBP

and SOE solutions.

Larger values of some indicators indicate the better performance (increasing indicators) while lower values of other indicators mean better performance (decreasing indicators). These can all be transformed into unified values as follows (Eq. (6) for increasing and Eq. (7) for decreasing indicators):

$$L(P_{i,j,k}) = \begin{cases} \frac{P_{i,j,k} - \min_{l=1}^{J_i}(P_{i,l,k})}{\max_{l=1}^{J_i}(P_{i,l,k}) - \min_{l=1}^{J_i}(P_{i,l,k})}, & \max_{l=1}^{J_i}(P_{i,l,k}) - \min_{l=1}^{J_i}(P_{i,l,k}) \neq 0; \\ 1, & \max_{l=1}^{J_i}(P_{i,l,k}) - \min_{l=1}^{J_i}(P_{i,l,k}) = 0 \end{cases} \quad (6)$$

$$L(P_{i,j,k}) = \begin{cases} \frac{\max_{l=1}^{J_i}(P_{i,l,k}) - P_{i,j,k}}{\max_{l=1}^{J_i}(P_{i,l,k}) - \min_{l=1}^{J_i}(P_{i,l,k})}, & \max_{l=1}^{J_i}(P_{i,l,k}) - \min_{l=1}^{J_i}(P_{i,l,k}) \neq 0; \\ 1, & \max_{l=1}^{J_i}(P_{i,l,k}) - \min_{l=1}^{J_i}(P_{i,l,k}) = 0 \end{cases} \quad (7)$$

$$L(P_{i,j}) = \frac{\sum_{k=1}^K w_k L(P_{i,j,k})}{\sum_{k=1}^K w_k} \quad (8)$$

All the SOBP performance indicators are first transformed by using Eqs. (6) and (7) and then the overall performance is calculated by using Eq. (8) with the corresponding weights.

After the performance of each SOBP scheme is obtained using Eq. (5) by the simulations, the scheme with maximum performance is selected as the best solution according to the assessment criterion.

Stage 3 Simulation-based SOEN performance

optimization

An SOEN is created when all the SOBP schemes in Stage 2 are connected together by services. Its performance optimization relates to all elements as shown in Figs. 1 and 2 for the entire SOEN system. The SOEN simulation model can be formulated and optimized with simulation software such as Arena and its OptQuest^[20]. The performance optimization is illustrated by a case SOEN in Section 4.

4 Case Study

Figure 3 illustrates a multi-echelon inventory system for a supply chain in a service-oriented computing environment. It contains K layers, each layer has some inventory nodes, and (j, k) denotes the j -th node of the k -th layer. Only the neighboring layers have supply-order relationships and the logistic direction is from the $(k+1)$ -th layer nodes to the k -th ones.

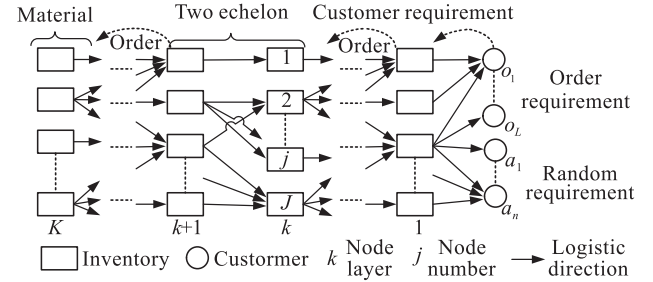


Fig. 3 Multi-echelon inventory system with SOEs

Suppose that some nodes are SOE. Their typical processes are (1) to order parts from the suppliers, (2) to produce well-configured products, and (3) to try to meet the next layer customers' needs. The SOBP has several ASVs which have some PSVs, e.g., the ASV named Transport2 has six candidate PSVs as shown on the right side of Fig. 4.

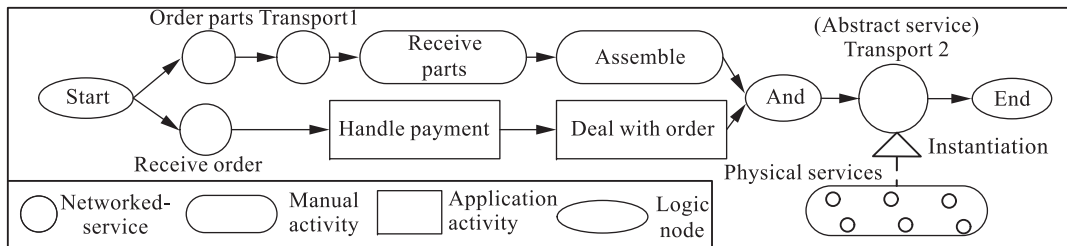


Fig. 4 Typical SOBP model of an SOE

4.1 Stage 1: Primary selection of PSV_{ij} for each ASV_i

Consider just the five performance indicators, R_b , T_b ,

C_b , T_q , and A_q of the activity named “Transport2”, i.e., the indicator set $P_i = \{R_b, T_b, C_b, T_q, A_q\}$. Let the fuzzy remark set $V = \{\text{good, medium, bad}\}$ and the judge matrix,

Table 4 Indicators for activities in Fig. 4

Activity	No.	R_b	T_b	C_b	F_b	O_b		
						S3	S4	S5
Order parts	1	0.90	30-50	200	4	0.213	0.177	0.142
Transport1	2	0.90	60-80	500	4	0.079	0.138	0.177
Receive parts	3	1.00	10-20	400	1	0.576	0.504	0.648
Assemble	4	1.00	200-260	4000	1	0.576	0.504	0.648
Receive order	5	1.00	30-80	300	1	0.576	0.504	0.648
Handle payment	6	1.00	60-80	100	1	0.576	0.504	0.648
Deal order	7	1.00	80-100	200	1	0.576	0.504	0.648
Transport2 (instantiated by S3, S4, or S5)	8	0.85	200-260	1000	6	0.059	—	—
	9	0.95	300-380	1200	8	—	0.213	—
	10	0.90	300-380	1400	10	—	—	0.213

Table 5 Simulation outcomes for SOBP schemes

Test No.	Process cycle time (T_b)			Resource utilization		
	S3	S4	S5	S3	S4	S5
1	6475.2	6398.9	6342.5	4.369	4.361	4.536
2	6751.6	6826.5	6613.7	4.224	4.375	4.356
3	6802.2	6719.7	6525.7	4.203	4.280	4.332
4	6467.9	6594.4	6875.3	4.379	4.320	4.292
5	6428.5	6142.8	6528.5	4.462	4.481	4.433
6	6563.3	6626.6	6671.4	4.386	4.287	4.359
7	6884.9	6544.2	6820.3	4.183	4.390	4.194
8	6450.3	6870.9	6569.4	4.454	4.184	4.403
9	6542.1	6317.9	6727.3	4.438	4.477	4.387
10	6721.9	6595.3	6379.2	4.324	4.260	4.476

The business reliabilities, flexibilities, organizational relations, business times, costs, and resource utilizations for the SOBP schemes are listed in Table 6.

Then, the AHP method gives the weight set (0.1348, 0.0802, 0.2107, 0.2345, 0.2431, 0.0967) for the indicators in Table 6. Equations (6)–(8) then give the overall performance of the SOBP schemes as

$$(P_1, P_2, P_3) = (0.0642, 0.1089, 0.1385).$$

Therefore, the overall performance order is $P_3(S5) > P_2(S4) > P_1(S3)$ and scheme 3 has the optimal overall performance for these criteria.

4.3 Stage 3: SOEN performance optimization

After the determination of each node's SOBP scheme,

the case SOEN inventory system in Fig. 3 is specified. Its performance can be optimized based on simulations by the reconfiguration of the inventories policies, such as the re-order policy with $(R_{j,k}, S_{j,k})$.

The basic data for the case inventory system shown in Table 7 contains 5 layers with 3, 2, 1, 4, and 1 nodes in each layer. Each node has $u = 1$ or 2 kinds of materials and $v = 1$ or 2 kinds of products, so their material-product quantity relations are $\lambda(u, v)$. The productivities of each node are $P(v^i)/t$, $u, v, i = 1$ or 2.

In an inventory system, some variables such as the costs (i.e., the products' unit cost, holding cost, production cost, shortage cost, re-order cost, and transport cost), time (e.g., the working schedule, re-order time

Table 6 Performance of different SOBP schemes

SOBP scheme	R_b	T_b	C_b	F_b	O_b	Resource utilization	Overall performance
Scheme 1 (S3)	0.6885	6608.8	6700	96	0.3319	4.34	0.0642
Scheme 2 (S4)	0.7695	6563.7	6900	128	0.3491	3.91	0.1089
Scheme 3 (S5)	0.7290	6605.3	7100	160	0.4214	4.38	0.1385

Table 7 Multi-echelon inventory system data

Node	(j,k)	u	v	$\lambda(1,1)$	$\lambda(2,1)$	$\lambda(1,2)$	$\lambda(2,2)$	$P(v^1)/t$	$P(v^2)/t$
1	(1,1)	1 or 2	1	1	1	0	0	10	0
2	(2,1)	1 or 2	2	0	0	2	1	0	10
3	(3,1)	1 or 2	1/2	1	1	2	1	10	10
4	(1,2)	1 or 2	1	1	2	0	0	50	0
5	(2,2)	1 or 2	2	0	0	2	2	0	30
6	(1,3)	1 or 2	1 or 2	2	1	2	2	100	160
7	(1,4)	2	1	1	2	0	0	250	0
8	(2,4)	1	2	0	0	3	1	0	160
9	(3,4)	1 or 2	1 or 2	1	2	3	1	250	160
10	(1,5)	1 or 2	1 or 2	1	1	1	1	1500	1200

point, inventory checking cycle), and policies (e.g., inventory control policies, reorder inventory level, order quantity) are controlled variables, while other variables are not controlled (e.g., customer's requirements and backorder delays). Most of them are usually not determinate but random variables.

Arena and OptQuest were used to optimize the performance of the case system for the performance operators defined in Section 3.3. The reorder policies and their parameters for the ten nodes' products, i.e., the $(R_{j,k,1}, S_{j,k,1})$ and $(R_{j,k,2}, S_{j,k,2})$, are set according to Table 7 in OptQuest, with the different combinatorial schemes set as the control factors. The cycle time for the customer orders (total time, t_t), the fulfillment rate of the orders (total fulfillment, f_t), and the total cost (c_t) of the system are the response factors. The simulation optimization minimizes c_t with the constraint of $f_t > 0.8$. The simulation is used to obtain the optimal policies solution, s^* , which is listed in Table 8. Then, s^* solution is run 10 times with Arena to obtain the optimal system

performance and their half-width (w_h) confidence levels: $t_t^* = 5.6547$, $w_{h,t} = 5.57$; $f_t^* = 0.92$, $w_{h,f} = 0.06$; and $c_t^* = 1\ 016\ 471$, $w_{h,c} = 38\ 893$.

5 Conclusions

Information technologies such as SOA and MDA have significantly influenced the IT and business environments of enterprises. Service-oriented enterprises and SOE networks are now emerging. This paper proposes a model-driven and service-oriented SOE architecture. Then, a three-stage simulation-based performance analysis and optimization method for service-oriented business process, SOE, and SOEN is developed where Stage 1 is an AHP-based overall performance analysis for service matching and binding, Stage 2 is a simulation-based performance evaluation for the business process/service composition, and Stage 3 is a business process simulation-based SOEN performance optimization. The combination of MDA, SOA, and PAO enhances the predictability, agility, flexibility, robustness, and measurability of the SOE and SOEN.

The MDA brings the services definition to a higher level of abstraction. A transformation is used to implement the services on a specific platform. The services and SOA are then decoupled from the lower level application platforms, IT infrastructures, and implementations, opening the way to improve enterprise integration in a service-oriented computing environment.

The three-stage simulation-based PAO method with the SOBP, SOE, and SOEN performance analysis framework, performance indicators, and performance operators ensures the correct service selection and service composition, and optimizes their performances.

Table 8 Optimal multi-layer decision solution (s^*) and its parameters for the case study inventory system

Node	(j,k)	$(R_{j,k,v}, S_{j,k,v})$	
		$v = 1$	$v = 2$
1	(1,1)	(30, 70)	(0, 0)
2	(2,1)	(0, 0)	(30, 70)
3	(3,1)	(30, 70)	(30, 70)
4	(1,2)	(150, 350)	(0, 0)
5	(2,2)	(0, 0)	(90, 420)
6	(1,3)	(300, 700)	(480, 1120)
7	(1,4)	(750, 1750)	(0, 0)
8	(2,4)	(0, 0)	(480, 1120)
9	(3,4)	(750, 1750)	(480, 1120)
10	(1,5)	(4500, 10 500)	(3600, 8400)

An SOEN case study of a multi-echelon inventory system indicates that the complementary approach using MDA, SOA, and PAO methods enables convenient reconfiguration to improve enterprise integration in a service-oriented computing environment.

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